Pablo Pena

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EDUCATION

University of Texas at Austin August 2024 – Expected Fall 2027
B.S. in Mechanical Engineering Honors, Robotics Minor GPA: 4.0
Relevant Coursework: Robot Mechanism Design, Mechatronics, Mechanics of Materials, Materials Engineering

SKILLS

CAD & Simulation: SolidWorks, Siemens NX, FEA, GD&T, Tolerance Stack-Up, NI Multisim
Manufacturing: Machine Shop (Drill Press, Bandsaw, Mill Maintenance), 3D Printing, Laser Cutting, SMT Soldering, Wire Harness Fabrication, CF/Kevlar Composite Layups
Programming: Python, MATLAB, C#, Arduino/Embedded, SQL, Java

PROFESSIONAL EXPERIENCE

BlackPearl Technology June 2026 – August 2026
Mechanical Engineering Intern *The Woodlands, TX*

- Designed the mechanical enclosure for a dual-PCB electromechanical DAQ device, working closely with electrical engineers to integrate BLE, LoRaWAN, and M2M hardware into a single ruggedized housing
- Engineered a torsional snap-fit mechanism enabling tool-free battery swaps, meeting drop protection (3 ft) and C1D1/Exd hazardous-area rating requirements for field deployment
- Took the design from CAD through prototype and into production, reducing component count by 67% and cutting assembly time by 60% through fixture and process improvements

NASA L'SPACE MCA June 2025 – September 2025
Mechanical Payload Engineer *Remote*

- Designed the mechanical architecture of a 250 kg Mars rover, developing a 6061 aluminum chassis and titanium rocker-bogie suspension for operation across extreme terrain and thermal environments
- Developed mechanical requirements and verification plans for 1,500 N static loads, 6,000 N shock loads, 2,000 Hz vibrations, and 45° inclines
- Engineered a precision 6-wheel rocker-bogie mobility system with independent wheel actuation and 4-wheel steering, enabling obstacle traversal up to 2× wheel diameter and continued operation after 2 wheel failures

PROJECTS & RESEARCH

Walking Delivery Robot — Jansen Linkage Leg Mechanism October 2026 – Present

- Designed a Jansen walking linkage using open-source kinematic simulation to generate the initial coupler path, modeled all links in SolidWorks, and 3D printed a PETG prototype to validate the physical mechanism against simulation
- Collaborated with a CS teammate on a Python gradient-based optimization that refined initial dimensions to hit 7 target stance points over a 70 mm stride, cutting foot-path RMS error by 75% while keeping all transmission angles above 40°

UT Austin – Yu Research Group August 2026 – Present
Undergraduate Research Assistant (PI: Guihua Yu, PhD) *Austin, TX*

- Synthesized 50+ batches of copper nanowires for a project developing Pd/Ag-doped Cu catalysts for electrochemical CO_2 reduction to ethylene
- Characterized nanowire morphology with SEM and confirmed Pd/Ag incorporation with EDS
- Observed HAADF-STEM images showing Z-contrast features from Pd and Ag dopants distributed in Cu nanowires

RC Plane — Texas Design Build Fly — 6th Place Internationally (2025) August 2024 – May 2026

- Hand-fabricated 4 fuselage structural prototypes using hybrid composite layups and vacuum bagging
- Fabricated carbon fiber/Kevlar landing gear with ±45° CF plies for torsional stiffness and 0° Kevlar plies for deflection resistance
- Flight-tested an 8G airframe structure carrying a 70 lb payload with no visible fuselage deformation